

LLM-Orchestrated Optical-Fibre ISAC Platform for Sensing-Event Monitoring and Selective Network Response

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Abstract *We propose an LLM-driven integrated communication-and-sensing system for automated QoT monitoring, sensing-event interpretation, vibration localization, and adaptive network response. Our use case demonstrates real-time vibration-event classification and localization for proactive link-risk assessment.*

Introduction

Deployed optical-fibre networks are increasingly viewed not only as high-capacity communication infrastructure, but also as a distributed sensing substrate[1, 2]. Recent advances in optical-fibre integrated sensing and communication (ISAC) show that in-service fibres can provide physical awareness of vibration events, environmental disturbances and route-level risks while continuing to carry communication traffic[3–5]. Such capability is attractive for operational networks, where railway activity, construction work, cable perturbation or other mechanical disturbances may occur along the same fibre routes that support live services. However, sensing awareness alone is not sufficient for network operation: each detected event must be interpreted in relation to service condition, operational context and possible response actions.

This makes optical-fibre ISAC an orchestration problem rather than a standalone sensing task. Sensing measurements, communication telemetry, topology information, historical event records and operation policies must be aligned in a consistent workflow that can distinguish routine events from actionable ones. Conventional fixed-threshold alarm pipelines are poorly suited to this task because they tend to generate non-actionable alarms and cannot translate high-level operator intent into context-aware monitoring, diagnosis and response procedures. Large language models (LLMs) provide a promising interface and control layer for intent understanding, tool invocation and explanation generation[6–8]. Nevertheless, for optical networks, an LLM should not replace domain algorithms or infer network states from language alone. Instead, it should act as a grounded orchestrator that calls validated optical-sensing, communication-diagnosis and network-control modules, and keeps decisions traceable through structured machine-readable evidence.

In this demo paper, we present an LLM-orchestrated optical-fibre ISAC platform for sensing-event monitoring and communication-aware network response over an active communication fibre. The platform follows a four-layer ar-

chitecture comprising an ISAC physical layer, a knowledge and data layer, an expert module pool and an LLM-based orchestration layer, which together support natural-language intent parsing, sensing-event monitoring, pattern recognition, localization, QoT assessment and report generation. In the live demonstration, after receiving a user instruction, the platform continuously monitors the fibre link and detects a vibration event. The LLM orchestrator invokes the expert module pool to analyse the event, identify it as a train-induced vibration, localize its position and assess its impact on communication performance. Since the measured BER/QoT remains unaffected, no network reconfiguration or urgent notification is triggered; instead, the event is recorded and an explanatory report is generated. This demonstrates a live intent-to-operation ISAC prototype that closes the loop between natural-language operator intent, real-time fibre sensing, expert-grounded event interpretation, communication-impact assessment and selective reporting, while providing a practical path towards LLM-assisted autonomous optical-network operation.

LLM-Orchestrated Optical-Fibre ISAC Platform

Figure 1 shows the proposed four-layer architecture. The platform is organized into a physical layer, a knowledge and data layer, an ISAC expert module pool, and an LLM-based ISAC orchestration layer.

1) ISAC physical layer: The bottom layer consists of an active optical fibre link with integrated communication and sensing capabilities. The communication subsystem supports traffic-bearing services, while the sensing subsystem continuously monitors physical disturbances along the deployed fibre. In this implementation, polarization-based sensing operates continuously for low-overhead monitoring, whereas DAS is activated only for high-resolution localization or event classification. Communication telemetry and sensing measurements are jointly streamed to the data layer.

2) Knowledge and data layer: The second layer maintains the operational evidence required for

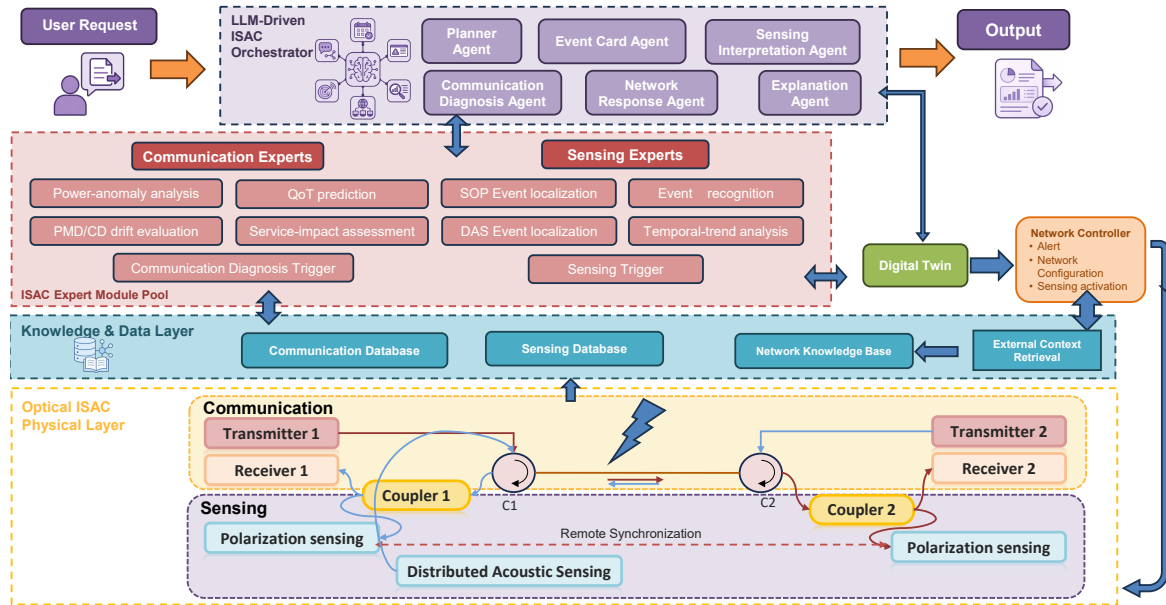


Fig. 1: LLM-driven optical-fibre ISAC Frame.

grounded orchestration, including coherent communication telemetry, polarization and DAS sensing data, topology information, link configurations, and historical event records. It also stores predefined KPIs and policies, enabling the LLM-based orchestrator to assess network conditions, support event interpretation, and generate appropriate control decisions.

3) ISAC expert module pool: The third layer contains pre-developed optical-domain algorithms exposed as callable expert modules. Sensing modules include polarization-event detection, SOP-based localization, DAS localization and event-pattern recognition. Communication modules include QoT estimation, power-anomaly analysis and service-impact evaluation.

4) LLM-based ISAC orchestration layer: The top layer connects the user interface with LLM agents for intent analysis and closed-loop orchestration. The Planner Agent parses the operator intent, while the Event Card Agent aggregates expert-module outputs into an LLM-readable card and transfers this structured state across agents. The Sensing Interpretation Agent and Communication Diagnosis Agent use domain knowledge to analyse the card and determine which expert modules should be invoked, and the Network Response Agent evaluates whether a control action is needed. Before a reconfiguration is applied, candidate actions are validated by a digital twin (DT). The validated command is then issued through the network controller to the physical layer, while the explanation agent generates a concise abnormal-event report for the user.

Use Case: LLM-Orchestrated Sensing Event Diagnosis

Figure 2 illustrates the workflow of the use case, in which an operator-level request is mapped to

continuous fibre monitoring, abnormal-event diagnosis, communication-impact assessment, QoT-aware validation, and final reporting.

Starting from the input intent shown at the top of Fig. 2, the Planner Agent parses the operator instruction into a monitor–diagnose–respond task chain and invokes the Sensing Trigger from the ISAC expert module pool. The platform remains in continuous monitoring mode without operator notification unless an abnormal sensing pattern or potential communication degradation is detected. In that case, the LLM orchestrator identifies an ISAC event and invokes the Event Card Agent to initialize the corresponding event card.

The event card records the event source, time window, sensing evidence, communication snapshot, BER status, confidence level, candidate diagnosis, and response state. This structured representation grounds LLM agents in machine-readable evidence. Based on the abnormality source and communication impact, the orchestrator invokes the Sensing Interpretation Agent for disturbance localization or pattern recognition, the Communication Diagnosis Agent for QoT degradation analysis, or both when physical disturbances are associated with communication risks.

These two LLM-based domain agents access optical-network knowledge and callable expert modules, but do not perform language-only inference or directly access raw physical-layer data. Instead, they ground the diagnosis on the current event card and invoke appropriate pre-trained or rule-based modules from the ISAC expert module pool. The sensing branch supports polarization-event interpretation, DAS-based localization, and vibration-pattern recognition, while the communication branch supports BER/Q-factor assessment, QoT prediction, power-anomaly analysis,

Acknowledgement

This work was partly supported by the European Commission's Horizon research and innovation program: Allegro project (No. 101092766) and the UK EPSRC project HASC (No. EP/X040569/1).

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